Cryptography and Security Mechanisms

Lecture 4A – Secret Key Cryptography
Data Encryption Standard

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Agenda

- **Block cipher principles**
- **DES**

Block Cipher Principles

Stream Ciphers and Block Ciphers

- Stream cipher: one bit or byte at a time
- Block cipher: a large block, typically 64 or 128 bits, at a time
  - Block ciphers work on a block at a time, which is some number of bits. All of these bits have to be available before they can be processed
  - Broader range of applications than stream ciphers
  - Large block thwarts statistical attacks

Block Cipher Principles

- **Motivation for the Feistel Cipher Structure**
  - **What is the most general form of block cipher?**
    - **For n-bit block, consider the transformation, F: $2^n \rightarrow 2^n$**
    - **F must be reversible, i.e., 1-1 correspondence**

Reversible Mapping	
Plaintext	**Ciphertext**
00	11
01	10
10	00
11	01

Irreversible Mapping	
Plaintext	**Ciphertext**
00	11
01	10
10	01
11	01

Ideal Block Cipher

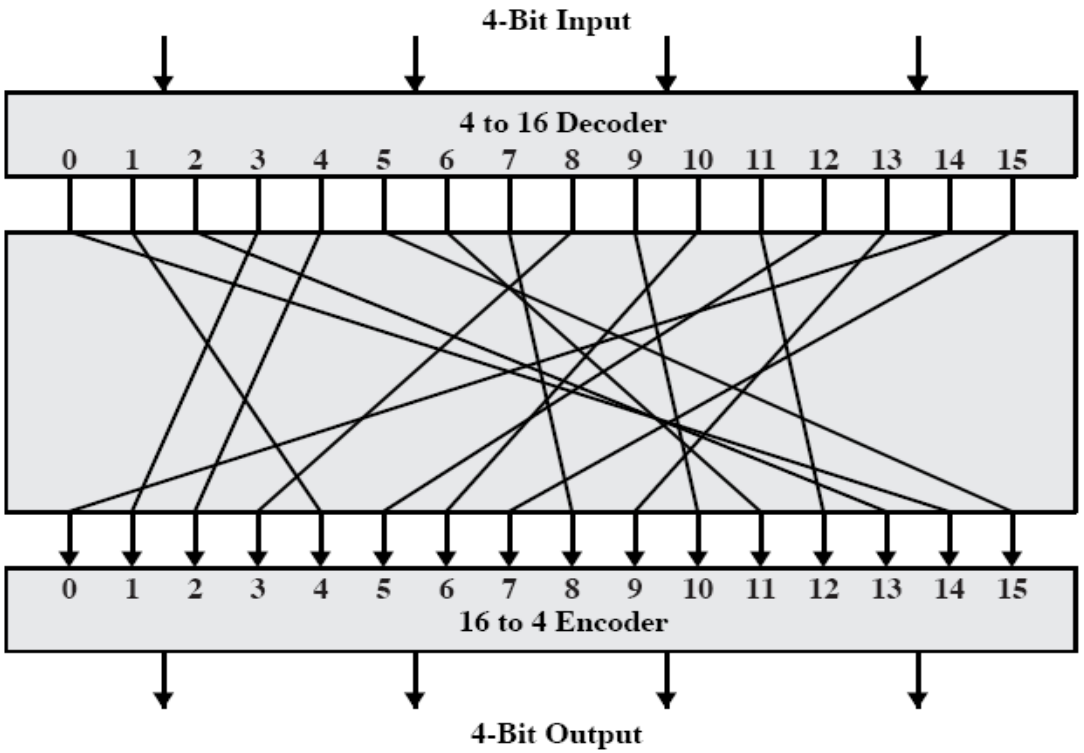


Figure 3.1 General *n*-bit-*n*-bit Block Substitution (shown with $n = 4$)

Plaintext	Ciphertext
0000	1110
0001	0100
0010	1101
0011	0001
0100	0010
0101	1111
0110	1011
0111	1000
1000	0011
1001	1010
1010	0110
1011	1100
1100	0101
1101	1001
1110	0000
1111	0111

Why ideal?

1. **Ideal bcz of unique plaintext-ciphertext mappings**
2. **Ideal bcz you can define as many as possible mappings ($2^n!$)**

Ideal Block Cipher

What if n is small?
What if n is large?

1. If n is small, system is vulnerable to statistical cryptanalysis
2. If n is large, statistical analysis for a large block is not feasible, however not practical from the implementation point of view

Diffusion and Confusion

Claude Shannon, 1949

- Diffusion: *hides the relationship b/w ciphertext and plaintext*
  - Diffusion means the change of a character in the plaintext changes several characters of the ciphertext. Similarly the change of a character in the ciphertext changes several characters in the plaintext
  - Spread the statistics over a range of bits-Thwart frequency analysis
  - *Can be achieved by repeatedly performing some permutation followed by applying a function to that permutation*
  - Hill cipher has this property : p_{11} changes, c_{11}, c_{21} and c_{31} also change

$$\begin{pmatrix} c_{11} & c_{12} & c_{13} \\ c_{21} & c_{22} & c_{23} \\ c_{31} & c_{32} & c_{33} \end{pmatrix} = \begin{pmatrix} k_{11} & k_{12} & k_{13} \\ k_{21} & k_{22} & k_{23} \\ k_{31} & k_{32} & k_{33} \end{pmatrix} \begin{pmatrix} p_{11} & p_{12} & p_{13} \\ p_{21} & p_{22} & p_{23} \\ p_{31} & p_{32} & p_{33} \end{pmatrix} \quad \text{mod } 26$$

$$c_{11} = (k_{11}p_{11} + k_{12}p_{21} + k_{13}p_{31}) \text{ mod } 26$$
$$c_{21} = (k_{21}p_{11} + k_{22}p_{21} + k_{23}p_{31}) \text{ mod } 26$$
$$c_{31} = (k_{31}p_{11} + k_{32}p_{21} + k_{33}p_{31}) \text{ mod } 26$$

Diffusion and Confusion

Claude Shannon, 1949

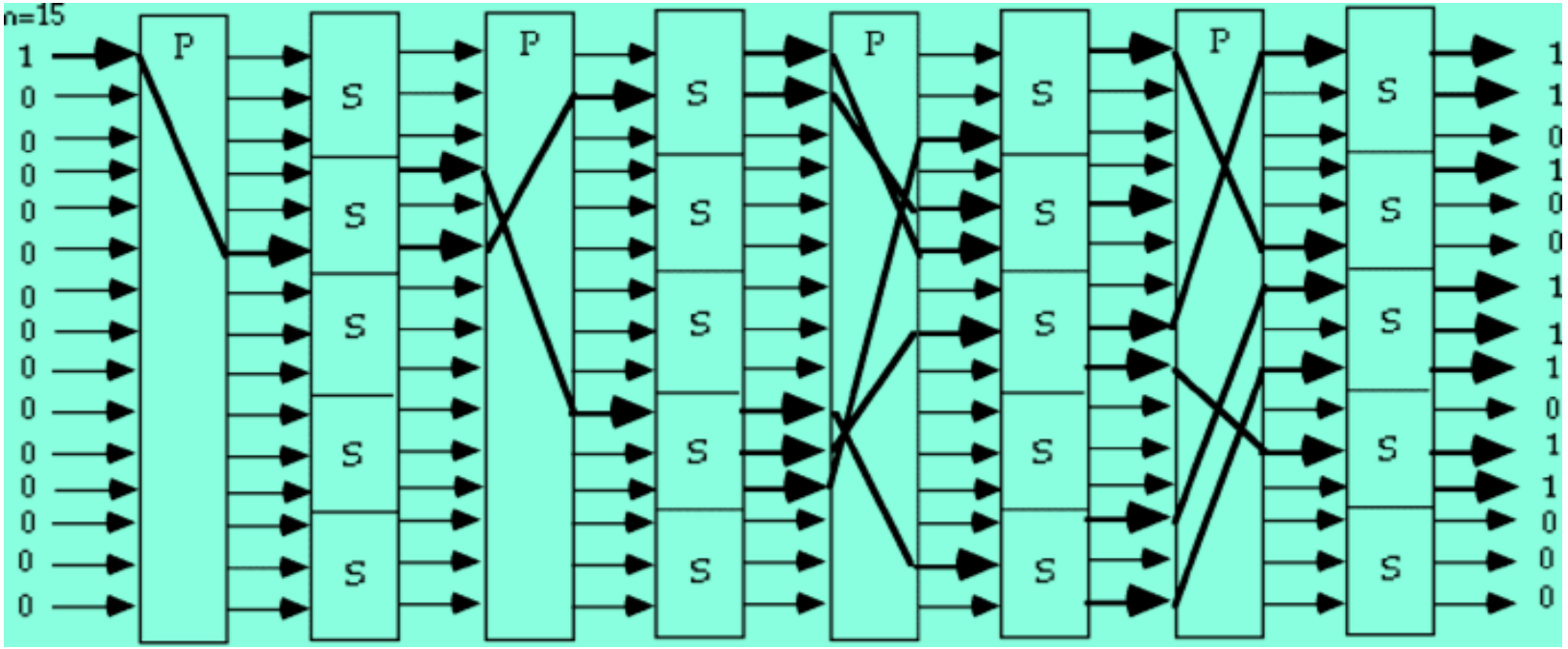
- Confusion: *hides the relationship b/w ciphertext and the key*
  - Confusion means that the key does not relate in a simple way to the ciphertext. In particular, each character in ciphertext should depend on several parts of the key
  - Make statistical relationship between the ciphertext and key as difficult as possible - Thwart attempts to discover the key
  - Can be achieved by using a complex, non-linear, substitution operation(S-box)
  - Hill cipher holds this property but not so good, k11 changes, changes the first row only i-e c_{11}, c_{12}, c_{13} it helps attacks to workout key partially.

$$\begin{pmatrix} c_{11} & c_{12} & c_{13} \\ c_{21} & c_{22} & c_{23} \\ c_{31} & c_{32} & c_{33} \end{pmatrix} = \begin{pmatrix} k_{11} & k_{12} & k_{13} \\ k_{21} & k_{22} & k_{23} \\ k_{31} & k_{32} & k_{33} \end{pmatrix} \begin{pmatrix} p_{11} & p_{12} & p_{13} \\ p_{21} & p_{22} & p_{23} \\ p_{31} & p_{32} & p_{33} \end{pmatrix} \quad \text{mod } 26$$

$$c_{11} = (k_{11}p_{11} + k_{12}p_{21} + k_{13}p_{31}) \text{ mod } 26$$
$$c_{12} = (k_{11}p1_{2} + k_{12}p_{22} + k_{13}p_{32}) \text{ mod } 26$$
$$c_{13} = (k_{11}p_{13} + k_{12}p_{23} + k_{13}p_{33}) \text{ mod } 26$$

Shannon Substitution-Permutation Network (S-P Network)

- **In 1949, Claude Shannon introduced the idea of substitution-permutation (S-P) networks which form the basis of modern block ciphers**
- S-P network: a special form of substitution-transposition product cipher
- Product cipher
  - Two or more simple ciphers are performed in sequence in such a way that the final result or product is cryptographically stronger than any of the component ciphers



Feistel Structure

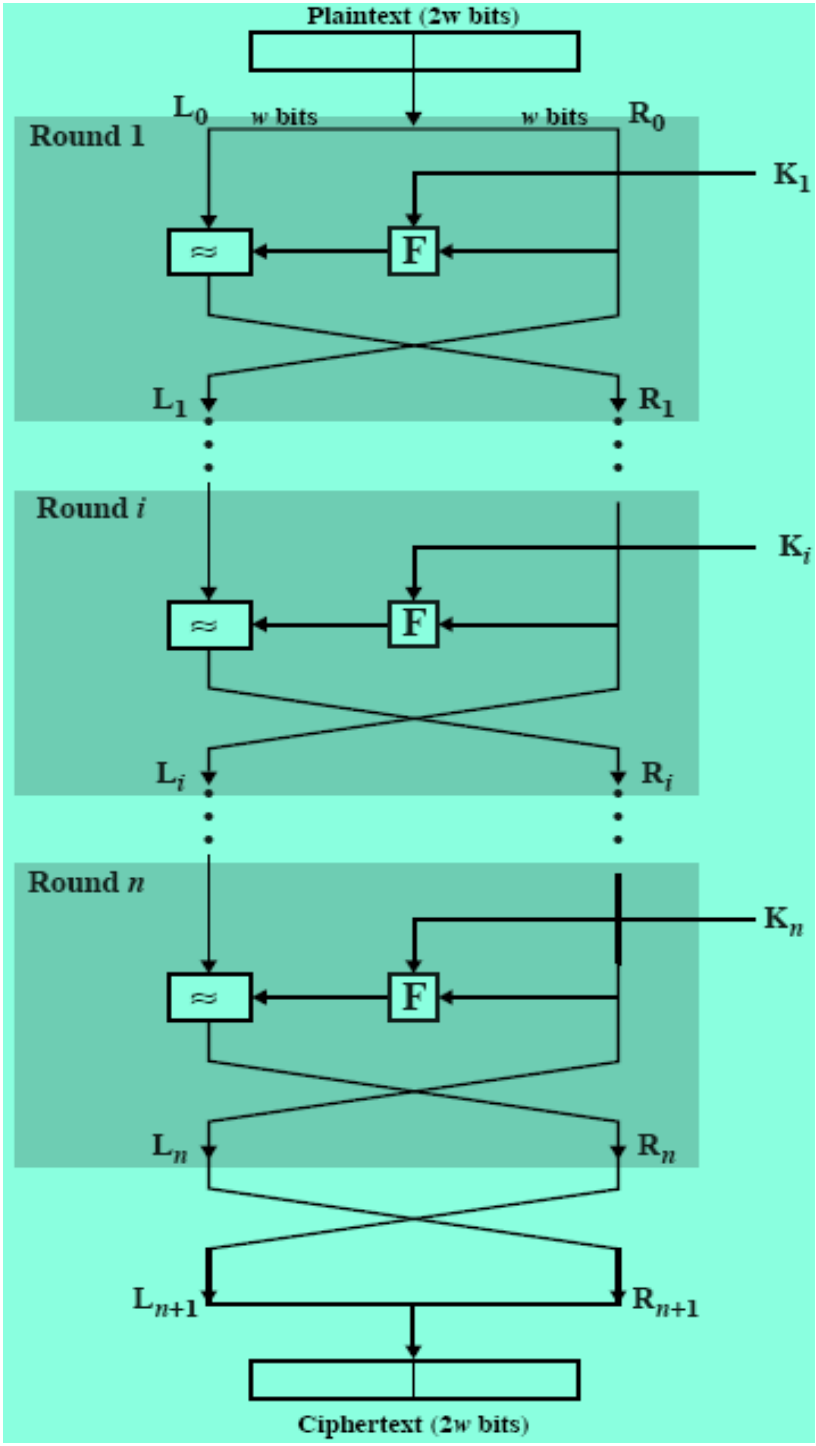
- Feistel cipher
  - In 1970's, Horst Feistel (IBM T.J. Watson Research Labs) invented a suitable (practical) structure which adapted Shannon's S-P network
  - The input is divided into two halves – the encryption function operates on one half and the second half is then mixed
  - Encryption and decryption use the same structure
  - The cipher has invertible, non-invertible and self-invertible components

Feistel Cipher Structure

- Input block is partitioned into two halves, L_{i-1} and R_{i-1}
- In round i,

  $L_i = R_{i-1}$

  $R_i = L_{i-1} \oplus F(K_i, R_{i-1})$
- Substitution followed by permutation
- Multiple rounds
- An Implementation of Shannon's S-P network (SPN) concept



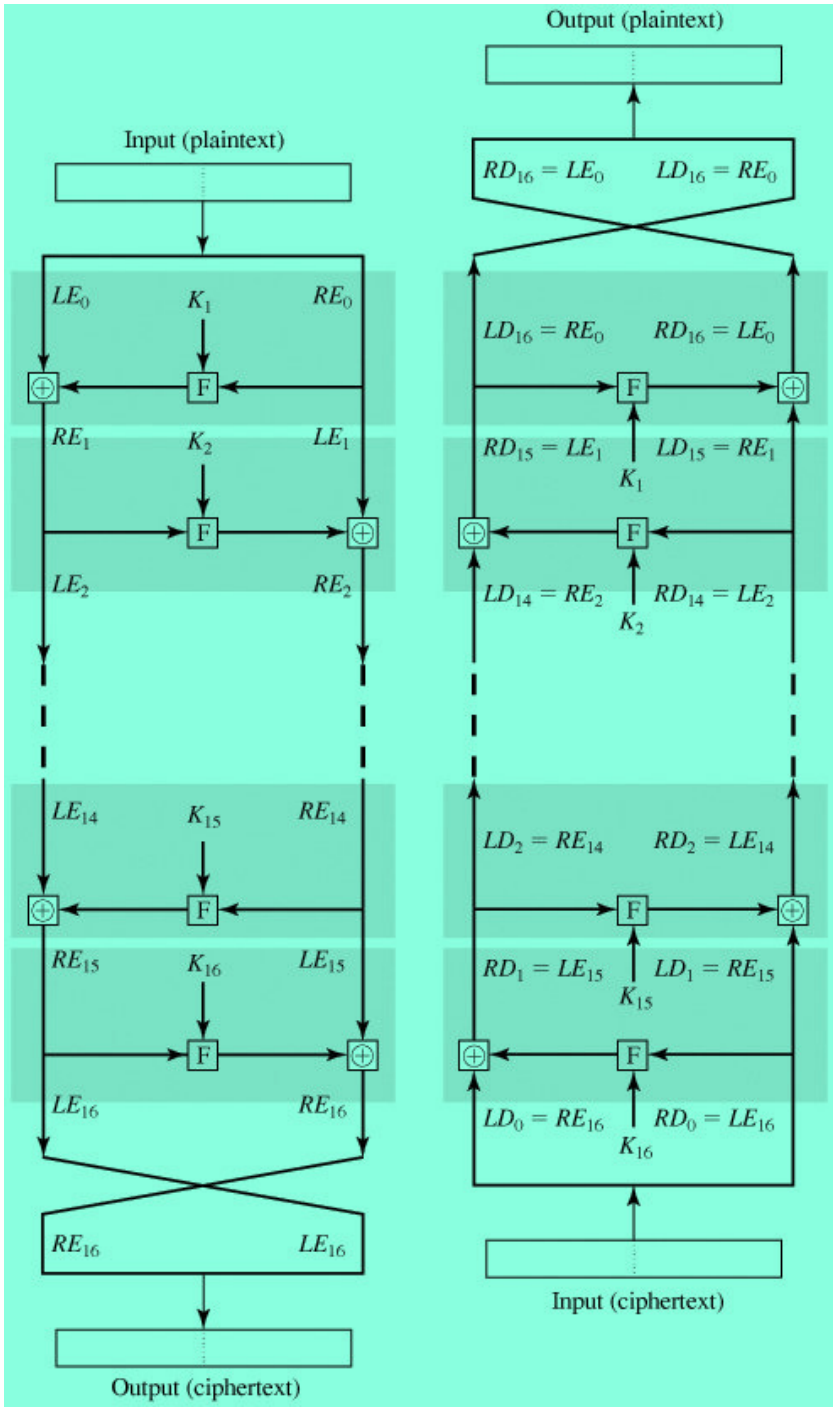
Feistel Cipher Design Elements

- Block size
  - Increasing size improves security, but slows cipher
- Key size
  - Increasing size improves security, makes exhaustive key searching harder, but may slow cipher
- Number of rounds
  - Increasing number improves security, but slows cipher
- Subkey generation algorithm
  - Greater complexity can make analysis harder, but slows cipher
- Round function
  - Greater complexity can make analysis harder, but slows cipher
- Fast software encryption/decryption
- Ease of analysis

Feistel Decryption Algorithm

- Essentially same as the encryption
- Same algorithm but use the subkeys K_i in reverse order
- F need not be a reversible function

- Satisfy $D_K(E_K(P)) = P$



Feistel Structure – Does decryption works?

On Encryption side

- **LE16=RE15**
- **RE16=LE15 x F(RE15, K16)**

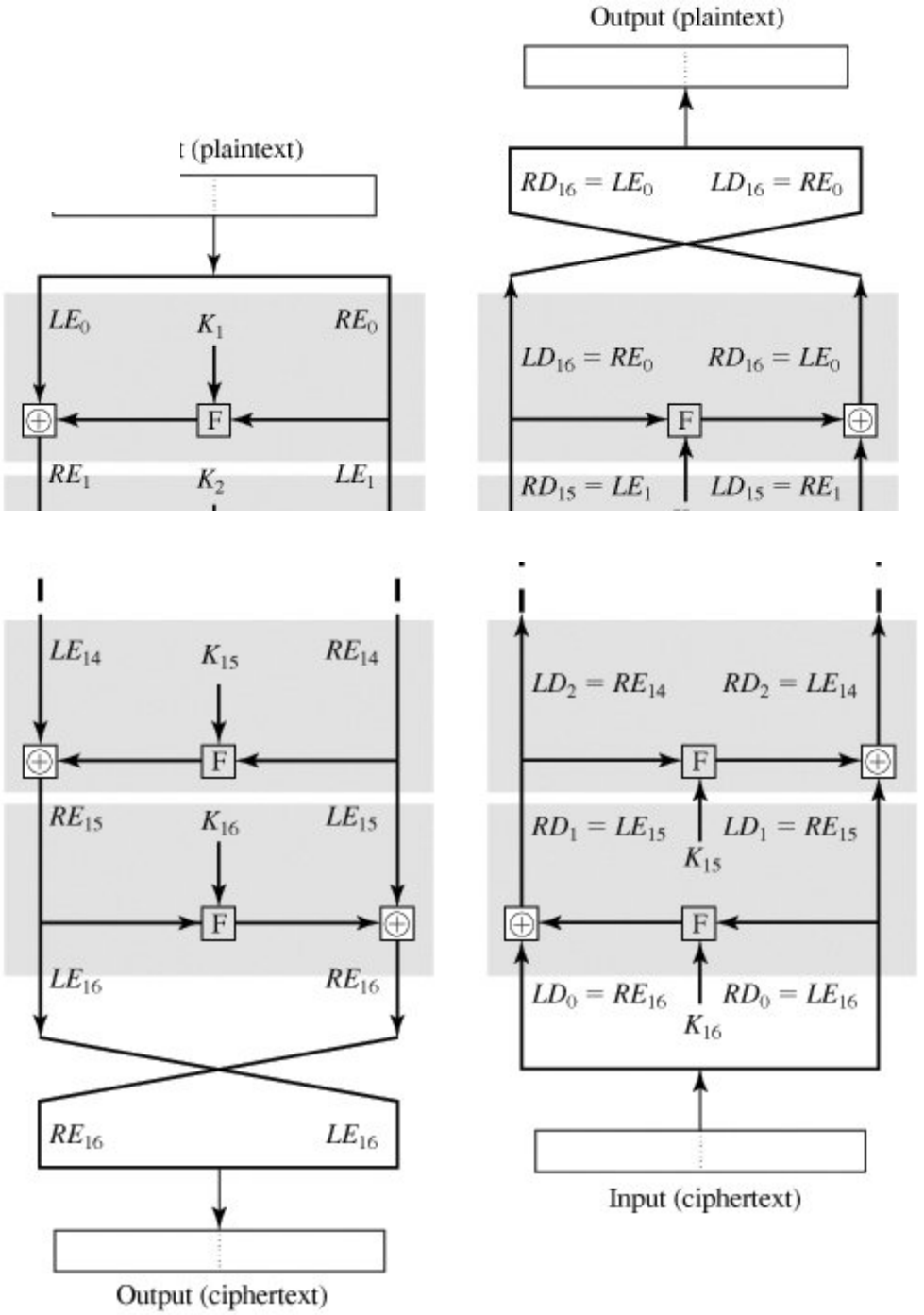
On decryption side

- **LD0=RE16 & RD0=LE16**
- LD1=RE15=RD0=LE16
- **RD1=LD0 x F(RD0, K16)**
- RD1=RE16x F(RE15, K16)
- RD1=LE15xF(RE15,K16)x F(RE15, K16)
- LE15

So

LD1=RE15 & RD1=LE15

That holds for all the 16 iterations



DATA ENCRYPTION STANDARD

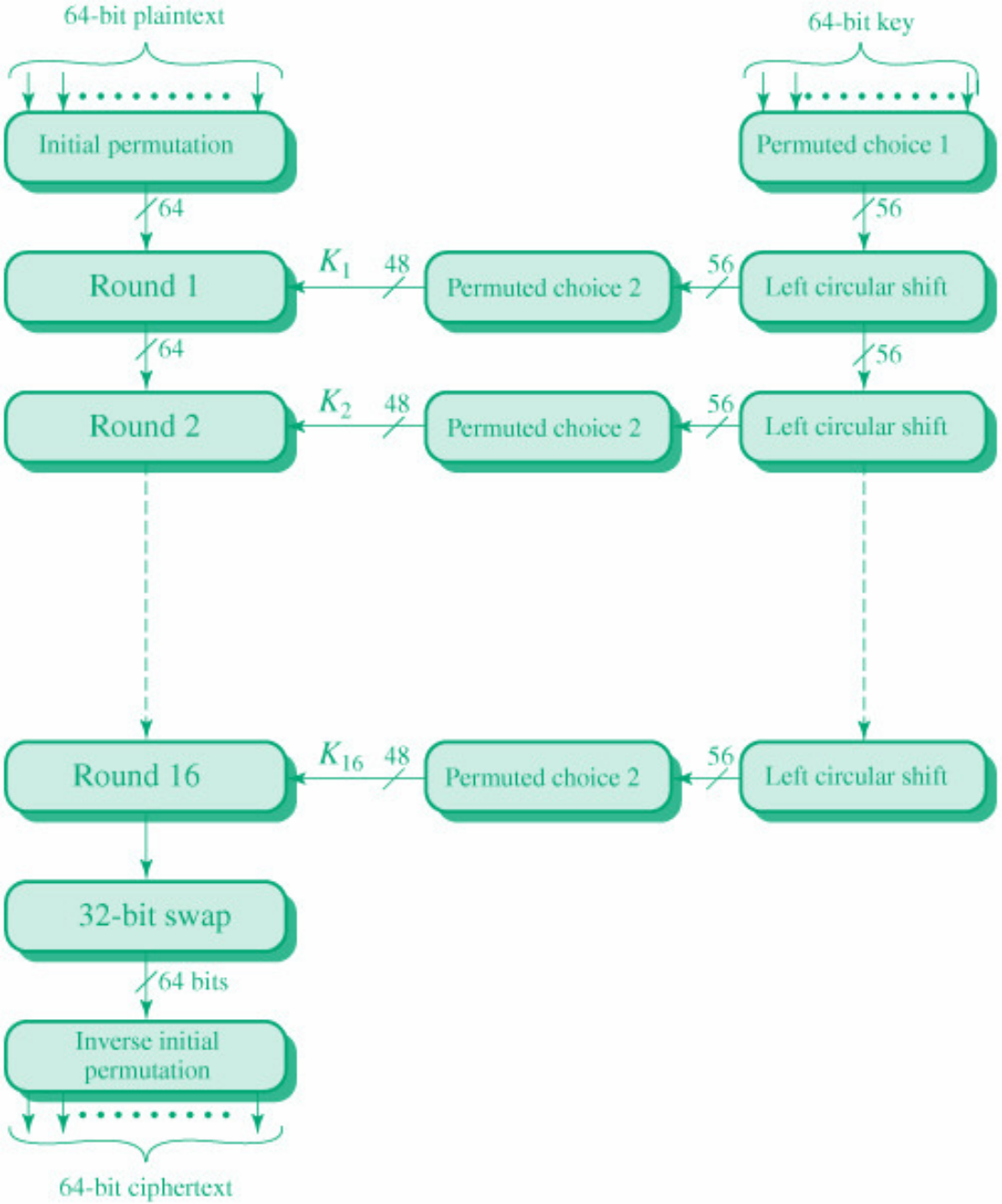
DES History

- In 1973, NBS (NIST) issues a public request for proposals for a national cipher standard, which must be
  - Secure
  - Public
  - Completely specified
  - Easy to understand
  - Available to all users
  - Economic and efficient in hardware
  - Able to be validated
  - Exportable
- IBM submitted LUCIFER (Feistel) (which was redesigned to become the DES).In 1977, adopted by NBS (NIST) as DES (Data Encryption Standard, Federal Information Processing Standard 46 (FIPS PUB 46))

DES Basics

- Block cipher (64-bit block)
- Uses 56-bit key (out of 64 bits)
- 16 rounds
- 48-bit subkeys
- Feistel structure (except initial and final permutations)

Overall Scheme of DES Encryption



- 64-bit input data goes thru initial permutation
- Then 16 rounds of the same iteration (round function is applied)
- For each round, sub-key is generated thru key generation module
- After 16 rounds of iterations, the contents of L and R are swapped and input to Inverse permutation
- Finally, a 64-bit ciphertext is done!

(details next..)

Input of DES

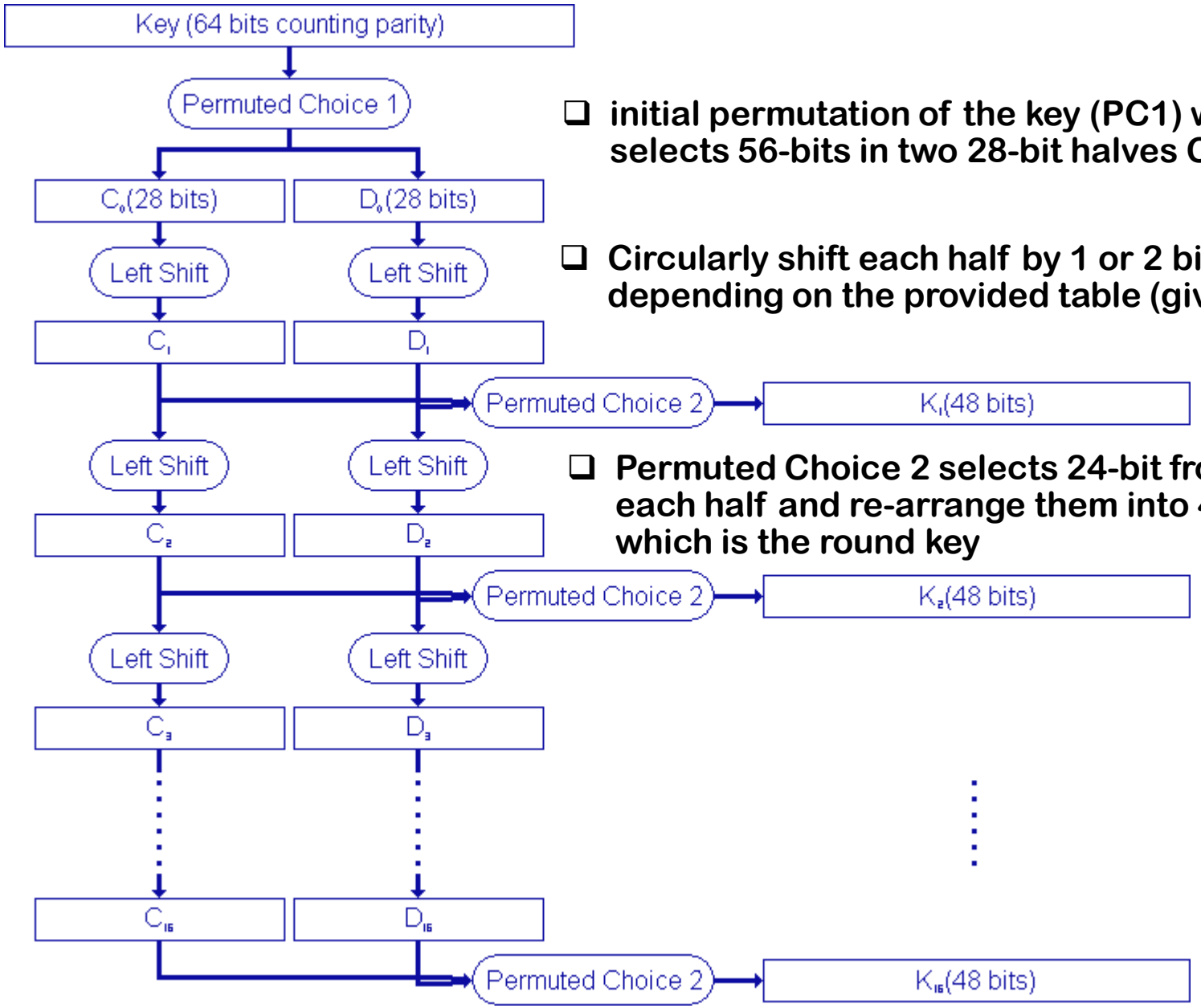
- **Data:** need to be broken into 64-bit blocks; add pad at the last message if necessary.
  - e.g. $X = (3\ 5\ 0\ 7\ 7\ F\ 1\ 0\ A\ B\ 1\ 2\ F\ C\ 6\ 5)_{HEX}$
- **Secret key:**
  - Any string of 64 bits long including 8 parity bits.
  - 1 parity bit in each 8-bit byte of the key may be utilized for error detection in key generation, distribution, and storage
  - $K=(k_1 \ldots k_7 k_8 \ldots k_{15} k_{16} k_{17} \ldots k_{24} \ldots k_{32} \ldots k_{40} \ldots k_{48} \ldots k_{56} \ldots k_{64})$
  - The bits $k_8, k_{16}, k_{24}, k_{32}, k_{40}, k_{48}, k_{56}, k_{64}$ can be used for parity check

DES KEY SCHEDULE

Generating subkeys used in each round

- consists of:
  - initial permutation of the key (PC1) which selects 56-bits in two 28-bit halves
  - 16 stages consisting of:
    - selecting 24-bits from each half
    - permuting them by PC2 for use in function f,
    - rotating each half separately either 1 or 2 places depending on the key rotation schedule K

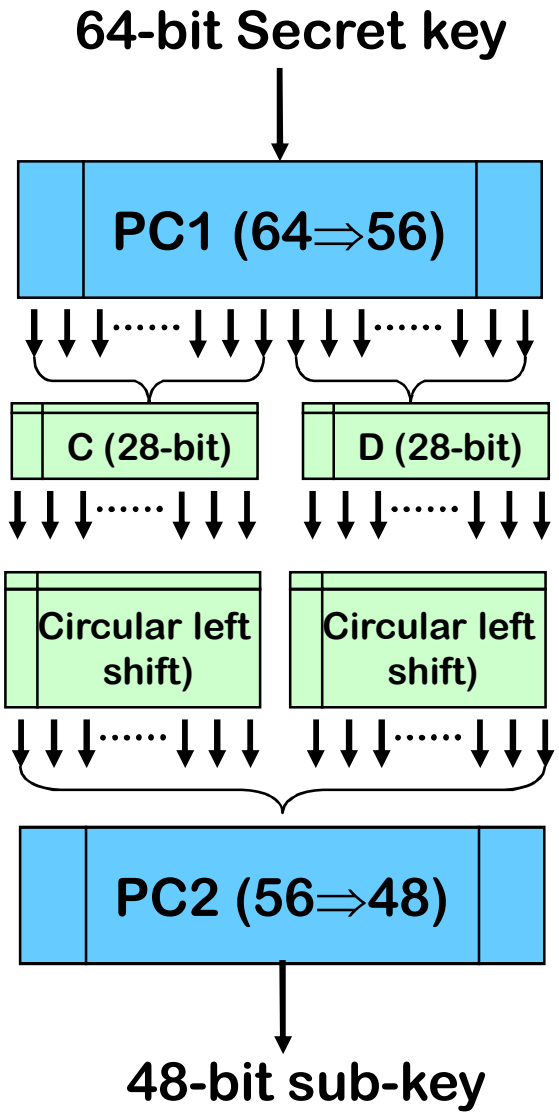
DES Key Schedule



- initial permutation of the key (PC1) which selects 56-bits in two 28-bit halves Ci & Di
- Circularly shift each half by 1 or 2 bit depending on the provided table (given)
- Permuted Choice 2 selects 24-bit from each half and re-arrange them into 48-bit which is the round key

One sub-key

- 64 bits of secret key are input to the key generator, 8 parity bits are removed; So, DES key has only 56 bits
- Objective: use these 56 bits to generate a different 48 bit sub-key for *each round* of DES
  - PC1 is a P box where 8 parity bits are removed with input of 64 bits key
  - 56-bit output of PC1 is split into two 28-bit keys which is input into shift registers C and D
  - The contents of are circularly shifted to left by 1 or 2 bits (according to a shift table) prior to each iteration
  - PC2 is also a P box which ignores certain input bits and permutes to a 48-bit sub-key



Sub-Key generations

- Given a secret key K of 64 bits long (includes 8 parity bits) by the sender
  - $K = [133457799BBCDFF1]_{16}$ $=[0001001100110100010101110111100110011011101111001101111111110001]_2$

Permuted Choice 1 (PC1)

- The table below specifies how the key is loaded to memory in PC1.

K=[0001001100110100010101111011110011001101110111100110111111110001]$_2$

K= [133457799BBCDFF1]$_{16}$ (64 bits)

Bit	57	49	41	33	25	17	9	1	58	50	42	34	26	18
Goes to bit	1	2	3	4	5	6	7	8	9	10	11	12	13	14
Bit	10	2	59	51	43	35	27	19	11	3	60	52	44	36
Goes to bit	15	16	17	18	19	20	21	22	23	24	25	26	27	28
Bit	63	55	47	39	31	23	15	7	62	54	46	38	30	22
Goes to bit	29	30	31	32	33	34	35	36	37	38	39	40	41	42
Bit	14	6	61	53	45	37	29	21	13	5	28	20	12	4
Goes to bit	43	44	45	46	47	48	49	50	51	52	53	54	55	56

PC1[k]= [11110000110011001010101011111010101010110011001111100011111]$_2$

= [F0CCAAF556678F]$_{16}$ (56 bits)

L = [1111000011001100101010101111]$_2$ = [F0CCAAF]$_{16}$ and

R = [0101010101100110011110001111]$_{2=}$ [556678F]$_{16}$ (28 bits each)

Shift Registers C and D

- The contents of $C = \{C_1, C_2, \ldots C_{28}\}$ and $D = \{D_1, D_2, \ldots D_{28}\}$ are circularly shifted to left by 1 or 2 bits (according to a shift table) prior to each iteration

$L = [11110000110011001010101011111]_{2} = [F0CCAAF]_{16}$
$R = [01010101011001100111100011111]_{2} = [556678F]_{16}$

- $C_1(L) = [11100001100110010101010111111] = [E19955F]_{16}$
  $D_1(R) = [10101010110011001111000111110] = [AACCF1E]_{16}$

Round	No. of Shift to left	Round	No. of Shift to left
1	1	9	1
2	1	10	2
3	2	11	2
4	2	12	2
5	2	13	2
6	2	14	2
7	2	15	2
8	2	16	1

Permuted Choice 2 (PC2)

- **PC2 is determined by the table below**
- **Consider input X= [C_1(L) D_1(R)] and Y=[C_2(L) D_2(R)]**
- $C_1(L)$ = [11100001100110010101010111111]= $[E19955F]_{16}$
  $D_1(R)$ = [10101010110011001111100011110] =$[AACCF1E]_{16}$
- X= [C_1(L) D_1(R)]= [11100001100110010101010111111
  10101010110011001111100011110]= $[E19955FAACCF1E]_{16}$
- PC2(X)= [00011011000000101110111111111110001110000011100 10]
  = $[1B02EFFC7072]_{16}$

Bit	14	17	11	24	1	5	3	28	15	6	21	10
Goes to bit	1	2	3	4	5	6	7	8	9	10	11	12
Bit	23	19	12	4	26	8	16	17	27	20	13	2
Goes to bit	13	14	15	16	17	18	19	20	21	22	23	24
Bit	41	52	31	37	47	55	30	40	51	45	33	48
Goes to bit	25	26	27	28	29	30	31	32	33	34	35	36
Bit	44	49	39	56	34	53	46	42	50	36	29	32
Goes to bit	37	38	39	40	41	42	43	44	45	46	47	48

K_1= $[1B02EFFC7072]_{16}$

16 round keys

$K = [133457799BBCDFF1]_{16}$

$K_1 = [1B02EFFC7072]_{16}$
$K_2 = [79AED9DBC9E5]_{16}$
$K_3 = [55FC8A42CF99]_{16}$
$K_4 = [72ADD6DB351D]_{16}$
$K_5 = [7CEC07EB53A8]_{16}$
$K_6 = [63A53E507B2F]_{16}$
$K_7 = [EC84B7F618BC]_{16}$
$K_8 = [F78A3AC13BFB]_{16}$
$K_9 = [E0DBEBEDE781]_{16}$
$K_{10} = [B1F347BA464F]_{16}$
$K_{11} = [215FD3DED386]_{16}$
$K_{12} = [7571F59467E9]_{16}$
$K_{13} = [97C5D1FABA41]_{16}$
$K_{14} = [5F43B7F2E73A]_{16}$
$K_{15} = [BF918D3D3F0A]_{16}$
$K_{16} = [CB3D8B0E17F5]_{16}$

DES ENCRYPTION
INITIAL PERMUTATION

Initial Permutation (IP)

- IP is determined as the following table
- It occurs before round one
- Bits in the plaintext are moved to next location, e.g. bit 58 to bit 1, bit 50 to bit 2 and bit 42 to bit 3, etc

58	50	42	34	26	18	10	2
60	52	44	36	28	20	12	4
62	54	46	38	30	22	14	6
64	56	48	40	32	24	16	8
57	49	41	33	25	17	9	1
59	51	43	35	27	19	11	3
61	53	45	37	29	21	13	5
63	55	47	39	31	23	15	7

Initial Permutation (IP) -Example

M=[0000000100100011010001010110011111000100110101011111001101111101111] = [123456789ABCDEF]$_{HEX}$

IP(M) =[1100110000000000110011001111111 1101 1100 0001 1111 0001 0000 1111 0100]= [CC00CCFF 9 F0AAF0AA]$_{HEX}$

Thus

- L_0 = 11001100000000001100110011111111 = CC00CCFF 9$_{HEX}$
- R_0 = 1101 1100 0001 1111 0001 0000 1111 0100 = F0AAF0AA $_{HEX}$

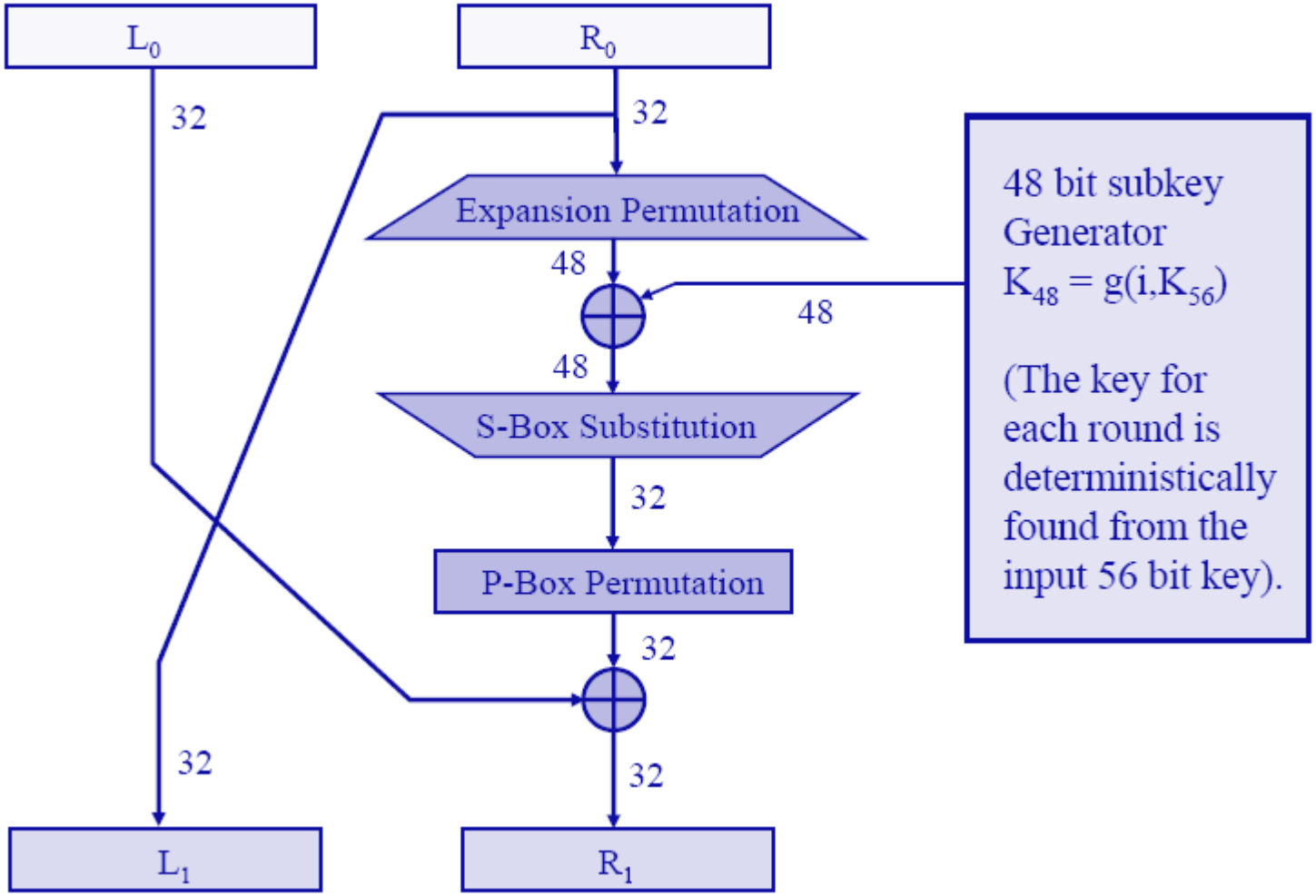
- L_0 and R_0 ready for iteration!

DES ENCRYPTION ROUND FUNCTION (SINGLE ROUND)

DES Round Function

- Operates on 32-bit units
- 32-bit $\rightarrow$ 48-bit expansion/permutation (E table)
- XOR with 48 bit subkey
- S-box computation returns 32 bits
- Round permutation (P)
- Followed by Feistel XOR and swap

Single Round of DES



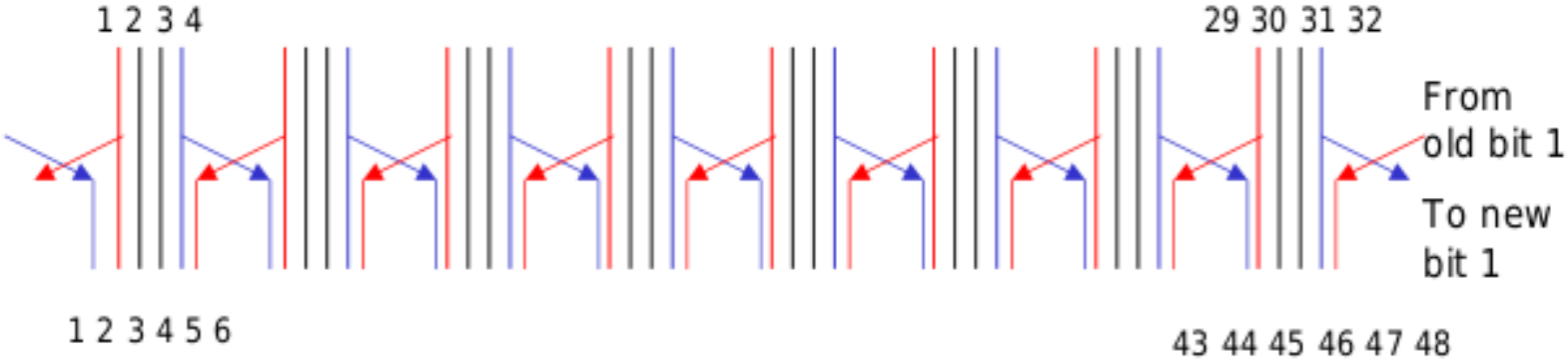
Computation of Round function $f(R_{i-1}, K_i)$

- Three types of boxes: E, S, P
- R (32 bits) is passed to expansion and permutation box E-box
- 48 bits output of E-box is added modulo 2 to 48 bits sub-key and result sent to S boxes
- S boxes (S_1, S_2…S_8) store a set of numbers; input 48 (=6×8) bits used to look up numbers like a code book and 32 bits output is sent to permutation box P
- Permutation box P permutes 32 bit input producing a 32-bit output

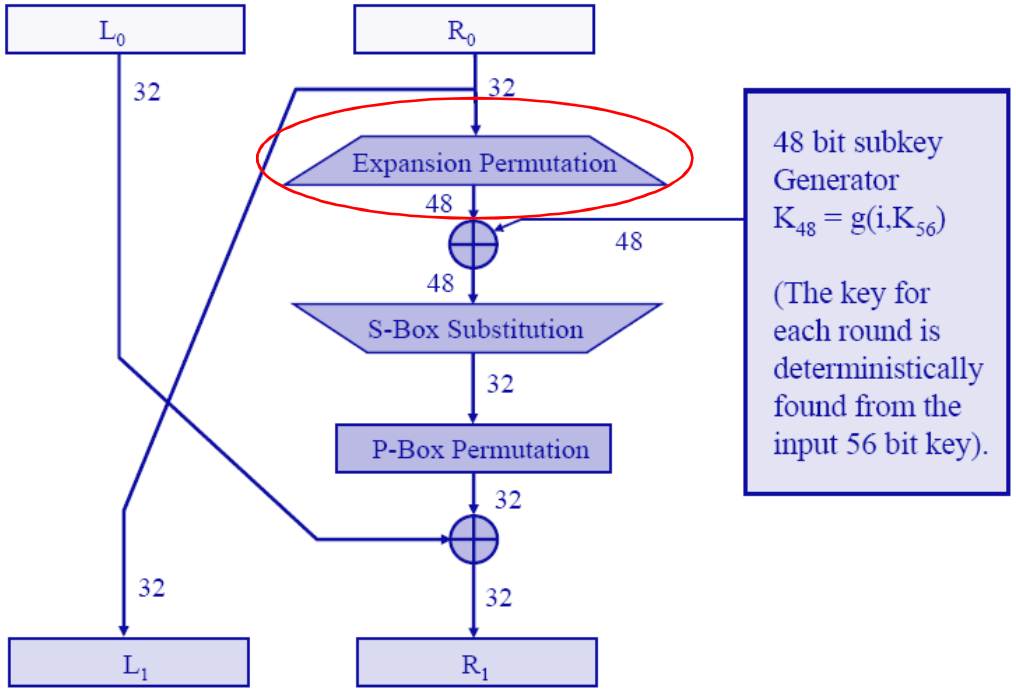
E-box used in DES

- The E-box expands 32 bits to 48 bits; it changes the order of the bits as well as repeating certain bits.

Bit	32	1	2	3	4	5	4	5	6	7	8	9
Goes to bit	1	2	3	4	5	6	7	8	9	10	11	12
Bit	**8**	**9**	**10**	**11**	**12**	**13**	**12**	**13**	**14**	**15**	**16**	**17**
Goes to bit	13	14	15	16	17	18	19	20	21	22	23	24
Bit	**16**	**17**	**18**	**19**	**20**	**21**	**20**	**21**	**22**	**23**	**24**	**25**
Goes to bit	25	26	27	28	29	30	31	32	33	34	35	36
Bit	**24**	**25**	**26**	**27**	**28**	**29**	**28**	**29**	**30**	**31**	**32**	**1**
Goes to bit	37	38	39	40	41	42	43	44	45	46	47	48

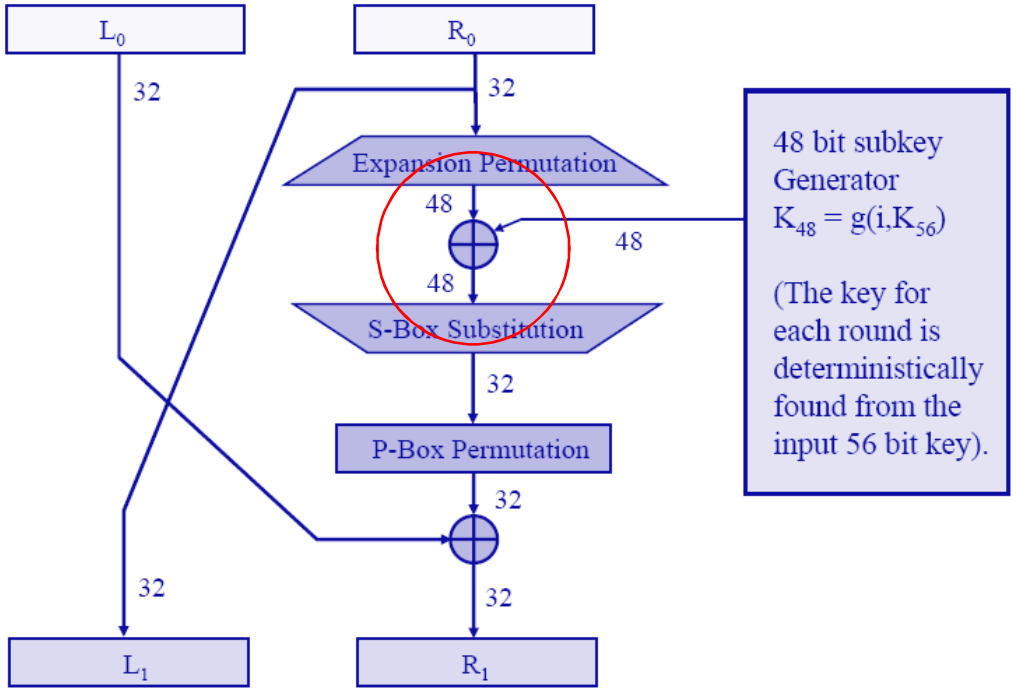


E-box used in DES



- L_0 = 11001100000000001100110011111111 = CC00CCFF 9_{HEX}
- R_0 = 1101 1100 0001 1111 0001 0000 1111 0100 = F0AAF0AA $_{HEX}$

E(R0) =[011110100001010101010101011110100001010101010101] = [7A15557A1555]$_{HEX}$



E(R0) =[01111010000101010101010101111010000101010101010101]

K_1 =[000110110000001011101111111110001110000011100010]

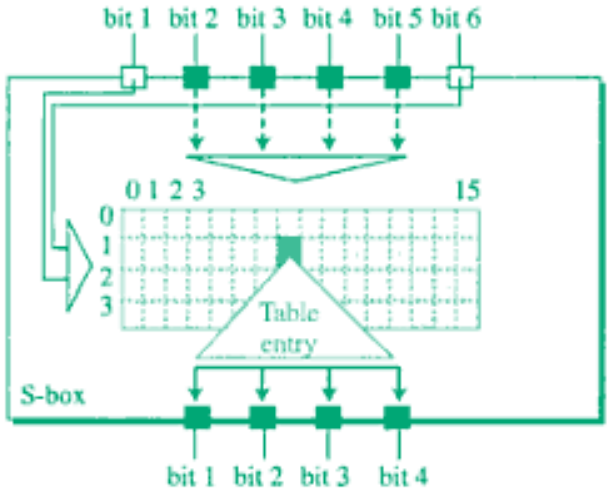
E(R0) ⊕ K1 =[011000010001011110111010100001100110010100100111]

=[6117BA866527]$_{HEX}$

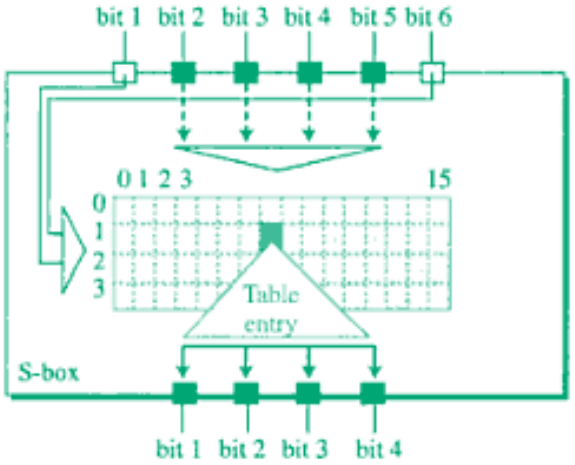
Substitution Boxes S

- Have eight S-boxes which map 6 to 4 bits
- Each S-box is actually 4 little 4 bit boxes
  - outer bits 1 & 6 (row bits) select one rows
  - inner bits 2-5 (col bits) are substituted
  - result is 8 lots of 4 bits, or 32 bits

- Example:
  S(1809123d11173839) = 5fd25e03



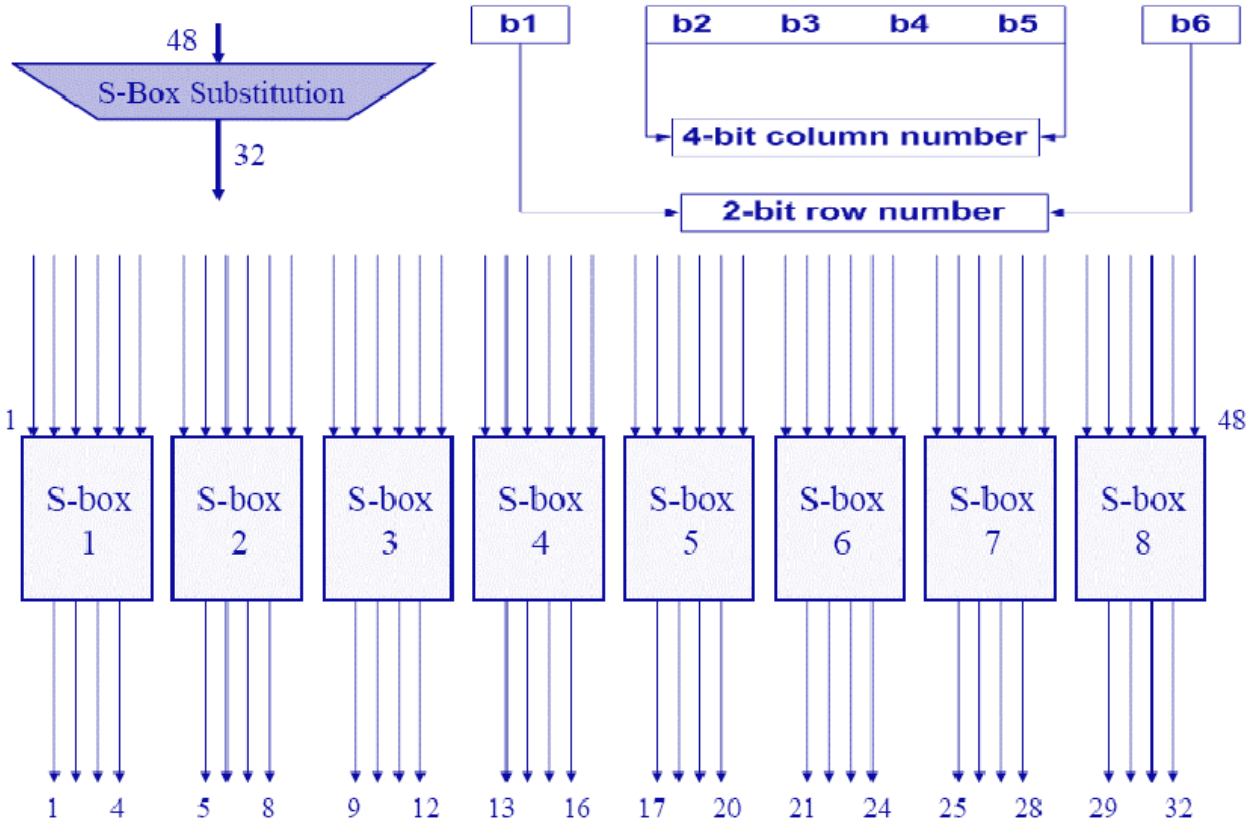
DES S-Boxes



שורה	מס' עמודה															
	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
	S_1															
0	14	4	13	1	2	15	11	8	3	10	6	12	5	9	0	7
1	0	15	7	3	14	2	13	1	10	6	12	11	9	5	3	8
2	4	1	14	8	13	6	2	11	15	12	9	7	13	10	5	0
3	15	12	8	2	4	9	1	7	5	11	3	14	10	0	6	13
	S_2															
0	15	1	8	14	6	11	3	4	9	7	2	13	12	0	5	10
1	3	13	4	7	15	2	8	14	12	0	1	10	6	9	11	5
2	0	14	7	11	10	4	13	1	5	8	12	6	9	3	2	15
3	13	8	10	1	3	15	4	2	11	6	7	12	0	5	14	9
	S_3															
0	10	0	9	14	6	3	15	5	1	13	12	7	11	4	2	8
1	13	7	0	9	3	4	6	10	2	8	5	14	12	11	15	1
2	13	6	4	9	8	15	3	0	11	1	2	12	5	10	14	7
3	1	10	13	0	6	9	8	7	4	15	14	3	11	5	2	12
	S_4															
0	7	13	14	3	0	6	9	10	1	2	8	5	11	12	4	15
1	13	8	11	5	6	15	0	3	4	7	2	12	1	10	14	9
2	10	6	9	0	12	11	7	13	15	1	3	14	5	2	8	4
3	3	15	0	6	10	1	13	8	9	4	5	11	12	7	2	14

שורה	מס' עמודה															
	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
	S_5															
0	2	12	4	1	7	10	11	6	8	5	3	15	13	0	14	9
1	14	11	2	12	4	7	13	1	5	0	15	10	3	9	8	6
2	4	2	1	11	10	13	7	8	15	9	12	5	6	3	0	14
3	11	8	12	7	1	14	2	13	6	15	0	9	10	4	5	3
	S_6															
0	12	1	10	15	9	2	6	8	0	13	3	4	14	7	5	11
1	10	15	4	2	7	12	9	5	6	1	13	14	0	11	3	8
2	9	14	15	5	2	8	12	3	7	0	4	10	1	13	11	6
3	4	3	2	12	9	5	15	10	11	14	1	7	6	0	8	13
	S_7															
0	4	11	2	14	15	0	8	13	3	12	9	7	5	10	6	1
1	13	0	11	7	4	9	1	10	14	3	5	12	2	15	8	6
2	1	4	11	13	12	3	7	14	10	15	6	8	0	5	9	2
3	6	11	13	8	1	4	10	7	9	5	0	15	14	2	3	12
	S_8															
0	13	2	8	4	6	15	11	1	10	9	3	14	5	0	12	7
1	1	15	13	8	10	3	7	4	12	5	6	11	0	14	9	2
2	7	11	4	1	9	12	14	2	0	6	10	13	15	3	5	8
3	2	1	14	7	4	10	8	13	15	12	9	0	3	5	6	11

DES S-Boxes



E(R0) ⊕ K1 =[011000010001011110111010100001100110010100100111]

=[6117BA866527]$_{HEX}$

S-Box[E(R0) ⊕ K1]

=[01011100100000101011010110010111]

=[5C82B597]$_{HEX}$

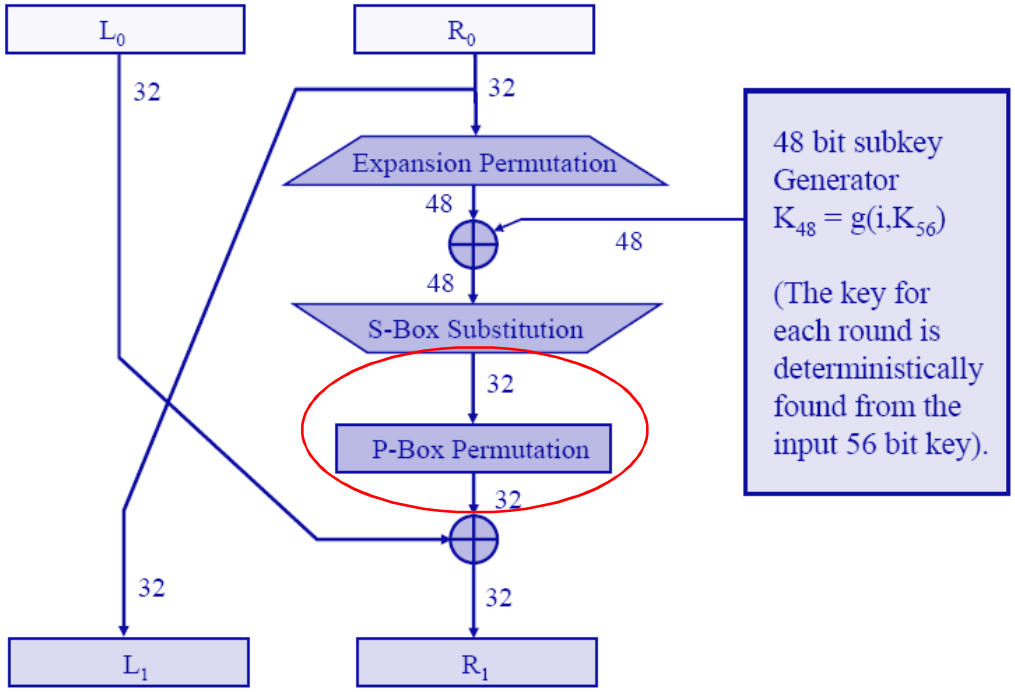
P-box used in DES

- The P-box permutation is determined as below which is a straight permutation; no bits are used twice, and no bits are ignored.

Bit	16	7	20	21	29	12	28	17	1	15	23	26	5	18	31	10
Goes to bit	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16

Bit	2	8	24	14	32	27	3	9	19	13	30	6	22	11	4	25
Goes to bit	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32

P-box used in DES



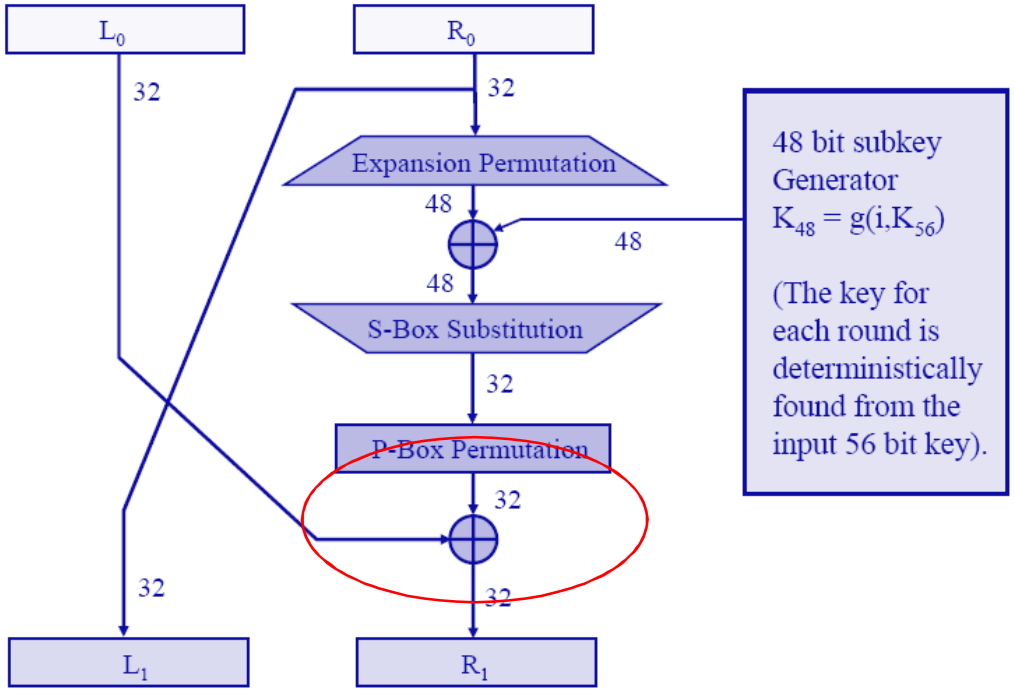
S-Box[E(R0) ⊕ K1]

=[010111001000001010110101100101 11] = $[5C82B597\ ]_{HEX}$

P[S-Box[E(R0) ⊕ K1]]

=[00100011010010101010100110111011] = $[234AA9BB\ ]_{HEX}$

P-box used in DES



P[S-Box[E(R0) ⊕ K1]]

=[00100011010010101010100110111011] = [234AA9BB]$_{HEX}$

L_0 = 11001100000000001100110011111111 = CC00CCFF 9$_{HEX}$

R_1= P[S-Box[E(R0) ⊕ K_1] ⊕ L_0

=[11101111010010100110010101000100] = [EF4A6544]$_{HEX}$

Round Outputs L_i , R_i

L_0 = 11001100000000001100110011111111 = [CC00CCFF 9]$_{HEX}$

R_0 = 1101 100 00011111 0001 0000 111 0100 = [F0AAF0AA] $_{HEX}$

L_1 = 1101 100 00011111 0001 0000 111 0100 = [F0AAF0AA]$_{HEX}$

R_1= 11101111010010100110010101000100 = [EF4A6544]$_{HEX}$

L_2 = 1101 100 00011111 0001 0000 111 0100 = [EF4A6544]$_{HEX}$

R_2= 11101111010010100110010101000100 = [CC017709]$_{HEX}$

L_3= 11101111010010100110010101000100 = [CC017709]$_{HEX}$

R_3= 10100010010111000000101111110100 = [A25C0BF4]$_{HEX}$

L_4 = 10100010010111000000101111110100 = [A25C0BF4]$_{HEX}$

R_4 = 01110111001000100000000001000101= [77220045]$_{HEX}$

L_5 = 01110111001000100000000001000101= [77220045]$_{HEX}$

R_5= 10001010010011111010011000110111 = [8A4FA637]$_{HEX}$

L_6= 10001010010011111010011000110111 = [8A4FA637]$_{HEX}$

R_6= 11101001011001111100110101101001 = [E967CD69]$_{HEX}$

L_7= 11101001011001111100110101101001 = [E967CD69]$_{HEX}$

R_7= 00000110010010101011101000010000 = [64ABA10]$_{HEX}$

L_8= 00000110010010101011101000010000 = [64ABA10]$_{HEX}$

R_8= 11010101011010010100101110010000 = [D5694B90]$_{HEX}$

Round Outputs L_i , R_i

L_9= 11010101011010010100101110010000 = $[D5694B90]_{HEX}$

R_9= 00100100011111001100011001111010 = [247CC67A] $_{HEX}$

L_{10}= 00100100011111001100011001111010 = [247CC67A] $_{HEX}$

R_{10} = 11101111010010100110010101000100 = $[B7D5D7B2\ ]_{HEX}$

L_{11} = 11101111010010100110010101000100 = $[B7D5D7B2\ ]_{HEX}$

R_{11}= 11000101011110000011110001111000= $[C5783C78\ ]_{HEX}$

L_{12}= 11000101011110000011110001111000= $[C5783C78\ ]_{HEX}$

R_{12}= 01110101101111010001100001011000 = $[75BD1858\ ]_{HEX}$

L_{13}= 01110101101111010001100001011000 = $[75BD1858\ ]_{HEX}$

R_{13} = 00011000110000110001010101011010= $[18C3155A]_{HEX}$

L_{14} = 00011000110000110001010101011010= $[18C3155A]_{HEX}$

R_{14}= 11000010100011001001011000001101 = $[C28C960D]_{HEX}$

L_{15}= 11000010100011001001011000001101 = $[C28C960D]_{HEX}$

R_{15}= 01000011010000100011001000110100 = $[43423234\ ]_{HEX}$

L_{16}= 01000011010000100011001000110100 = $[43423234\ ]_{HEX}$

R_{16}= 00001010010011001101100110010101= $[0A4CD995]_{HEX}$

Inverse Initial Permutation (IP^{-1})

- IP^{-1} is determined as the following table;
- Since DES consists of 16 rounds, too many for our lecture!
- Consider DES algorithm of two rounds.
- Ciphertext = $IP^{-1}(R_1L_1)$ = 1101 0111 0110 1001 1000 0010 0010 0100 0010 1000 0011 1110 0000 1010 1110 1010 = $D7698224283E0AEA_{HEX}$

40	8	48	16	56	24	64	32
39	7	47	15	55	23	63	31
38	6	46	14	54	22	62	30
37	5	45	13	53	21	61	29
36	4	44	12	52	20	60	28
35	3	43	11	51	19	59	27
34	2	42	10	50	18	58	26
33	1	41	9	49	17	57	25

Final Ciphertext

L_{16}= 01000011010000100011001000110100 = $[43423234\ ]_{HEX}$

R_{16}= 00001010010011001101100110010101 = $[0A4CD995]_{HEX}$

$IP^{-1}(R_{16}L_{16})$= IP^{-1}[00001010010011001101100110010101
01000011010000100011001000110100]

$IP^{-1}(R_{16}L_{16})$=
=[10000101111010000001001101010100000011110000101010110100000
00101= $[85E813540F0AB405\ ]_{HEX}$

Plaintext=[00000001001000110100010101100111100010011010
01011110011011110111]= $[123456789ABCDEF]_{HEX}$

Key=[0001001100110100010101110111100110011011101111010
0110111111110001] = $[133457799BBCDFF1]_{HEX}$

Ciphertext
=[10000101111010000001001101010100000011110000101010
1101000000101]= $[85E813540F0AB405\ ]_{HEX}$

DES Avalanche Effect

Table 3.5 Avalanche Effect in DES

(a) Change in Plaintext		(b) Change in Key	
Round	Number of bits that differ	Round	Number of bits that differ
0	1	0	0
1	6	1	2
2	21	2	14
3	35	3	28
4	39	4	32
5	34	5	30
6	32	6	32
7	31	7	35
8	29	8	34
9	42	9	40
10	44	10	38
11	32	11	31
12	30	12	33
13	30	13	28
14	26	14	26
15	29	15	34
16	34	16	35

- In any good cipher, any change in the key or plaintext, no matter how large or small, should change approximately half the ciphertext bits
- Examples
  - (a) Change one bit in the plaintext with the same key
  - (b) Change one bit in the key with the same plaintext
  - After 3 or 4 rounds, approximately half of the ciphertext bits are changed
  - After 16 rounds, a lot of scrambling has taken place

Home Work

How the same Feistel structure for DES encryption works for decryption ? Prove